

Irreducibility of exposed-point varieties of discotopes

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September 30, 2026 (version 1.0)

Abstract

We prove that the complex Zariski closure of the exposed points of any finite Minkowski sum of ellipsoidal discs of dimension at least two is irreducible. No genericity or full-dimensionality assumption is needed. This proves the exposed-point irreducibility conjecture stated by Meroni in the 2023 Oberwolfach report. Under an explicit generic condition on the summand spans, we also prove that this closure equals the purely nonlinear part defined by Gesmundo and Meroni, thereby proving their 2022 irreducibility conjecture. The proof uses a connected real-analytic support-point parametrization. A nongeneric example shows why the two varieties must be distinguished.

Status. Unrefereed preprint. Generative AI was used extensively in developing and checking the result and preparing this manuscript; no human peer review or formal proof certification is claimed.

1 The conjectures and the result

A *disc* is a linear image of a closed Euclidean unit ball. Thus it is an ellipsoidal ball in its linear span and is centered at the origin. A *discotope* is a finite Minkowski sum $D = D_1 + \cdots + D_N \subset \mathbb{R}^d$, with $N \geq 1$. A point of a compact convex set K is *exposed* if it uniquely maximizes some nonzero real linear functional on K . Write

$$E(K) = \overline{\text{Exp}(K)}^{\text{Zar}, \mathbb{C}} \subset \mathbb{C}^d.$$

Meroni's Conjecture 1 in [3, p. 830] asks whether $E(D)$ is irreducible for generic discotopes whose summands have dimension at least two. We prove the following stronger statement.

Theorem 1. *Let $D = D_1 + \cdots + D_N \subset \mathbb{R}^d$ be a discotope, with every D_i of dimension at least two. Then $E(D)$ is irreducible over $\mathbb{C}$, without genericity or full-dimensionality assumptions.*

Gesmundo and Meroni proved irreducibility for generic discotopes in the range $\sum_i (\dim D_i - 1) \leq d - 1$ (Theorem 4.3), and for generic sums of two-dimensional discs (Theorems 4.3 and 6.1). Remark 3.4 treats sums whose summands are all full-dimensional. Their Section 5 already uses the support-gradient parametrization. The contribution here is the general irreducibility conclusion for arbitrary summand ranks at least two, together with the generic identification of the two varieties. The proof uses elementary analytic and convex facts rather than an asserted irreducibility theorem for a complex critical locus.

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There is a distinction between this target and the earlier formulation. Gesmundo and Meroni define the *purely nonlinear part* [1, Definition 3.3] by

$$S(D) = \overline{D^\partial \cap \partial D}^{\text{Zar}, \mathbb{C}}, \quad D^\partial = \sum_i \partial_{L_i} D_i, \quad L_i = \text{span } D_i, \quad (1)$$

where each summand boundary is relative to its own span. Their Conjecture 8.2 asks for $S(D)$ to be irreducible for generic discotopes with no one-dimensional summands. In Section 3 we prove $S(D) = E(D)$ under an explicit condition holding generically, which proves that conjecture too. The two varieties can differ outside generic position, as Section 4 demonstrates. We do not address the separate questions about degrees, birationality, or the entire complex critical locus of the addition map.

2 A connected analytic parametrization

We first record two elementary facts. The analytic fact is a standard identity-theorem argument. Kucharz and Kurdyka explicitly record its real Zariski version in the proof of Proposition 4.3 of [2, p. 95]. We include the complex version for completeness.

Lemma 2. *The complement in $\mathbb{R}^d$ of a finite union of real linear subspaces of codimension at least two is nonempty, open, and path connected.*

Proof. A finite union of proper real subspaces cannot fill $\mathbb{R}^d$: choose a nonzero linear form vanishing on each subspace; their product is a nonzero polynomial. The complement is open because the union is closed. Let x, y be in the complement. If the excluded subspaces are M_i , each $M_i + \mathbb{R}x$ and $M_i + \mathbb{R}y$ has dimension at most $d - 1$. Choose z outside their finite union. If $(1 - t)x + tz \in M_i$ for $0 < t \leq 1$, then $z \in M_i + \mathbb{R}x$, a contradiction. The same holds for the segment from y to z . These two segments give the required path. $\square$

Lemma 3. *If $U \subset \mathbb{R}^d$ is nonempty, open, and connected and $F : U \rightarrow \mathbb{R}^m$ is real analytic, the complex Zariski closure of $F(U)$ is irreducible.*

Proof. Let $I \subset \mathbb{C}[X_1, \dots, X_m]$ be the ideal of polynomials vanishing on $F(U)$. It is proper. If $PQ \in I$ and $P \notin I$, there is a nonempty open subset of U on which $P \circ F \neq 0$, so $Q \circ F = 0$ there. The real-analytic identity theorem, applied to the real and imaginary parts on connected U , implies $Q \circ F = 0$ throughout U . Hence I is prime. It is exactly the vanishing ideal of the complex Zariski closure, so that closure is irreducible.

For completeness, analytic uniqueness here requires only connected openness. The points where a real-analytic function vanishes on a neighborhood form an open set. At a limit point in U , continuity of every derivative makes its Taylor coefficients zero; its convergent Taylor series makes the function zero on a neighborhood. Thus that set is also relatively closed, and connectedness completes the argument. $\square$

Proof of Theorem 1. Choose injective linear maps $A_i : \mathbb{R}^{m_i} \rightarrow \mathbb{R}^d$, with $m_i = \dim D_i \geq 2$, such that $D_i = A_i(B^{m_i})$, and put $Q_i = A_i A_i^T$. Such maps exist by restricting a singular-value decomposition to its nonzero singular directions. For a normal $u \in \mathbb{R}^d$, the support value of D_i is

$$h_i(u) = \|A_i^T u\| = \sqrt{u^T Q_i u}.$$

When $A_i^T u \neq 0$, equality in Cauchy–Schwarz gives the unique maximizing point

$$p_i(u) = \frac{A_i A_i^T u}{\|A_i^T u\|} = \frac{Q_i u}{\sqrt{u^T Q_i u}}. \quad (2)$$

When $A_i^T u = 0$, the entire positive-dimensional disc maximizes.

The face of a Minkowski sum exposed by u is the sum of the individual faces: support values add, and equality in their sum is equivalent to equality term by term. A sum of nonempty faces is a singleton exactly when each face is a singleton. Indeed, fixing a point in every other face translates any nonsingleton face injectively into the sum. Therefore, with

$$U = \mathbb{R}^d \setminus \bigcup_i \ker A_i^T, \quad F(u) = \sum_i \frac{Q_i u}{\sqrt{u^T Q_i u}}, \quad (3)$$

we have the exact set equality $\text{Exp}(D) = F(U)$. The origin is excluded from U , so every normal used is nonzero.

Every excluded kernel has codimension $m_i \geq 2$. Lemma 2 makes U connected and open. On U , every radicand is $\|A_i^T u\|^2 > 0$. The positive real inverse square root is real analytic on $(0, \infty)$, so F is real analytic on all of U . Lemma 3 now proves the assertion. $\square$

This argument allows repeated discs, coincident spans, dependent quadratic radicals, noninjective support parametrizations, and lower-dimensional sums. It neither chooses a global complex square-root branch nor assumes a rank condition on the differential of F .

3 The generic purely nonlinear part

Theorem 4. *Suppose D is full-dimensional, every D_i has dimension at least two, and its summand spans satisfy*

$$\dim\left(\sum_{i \in J} L_i\right) = \min\left(d, \sum_{i \in J} \dim L_i\right) \quad \text{for every } J \subseteq \{1, \dots, N\}. \quad (\text{GP})$$

Then $S(D) = E(D)$; in particular $S(D)$ is irreducible over $\mathbb{C}$.

Proof. Every exposed point is a sum of the points (2), each on its relative summand boundary, so $E(D) \subseteq S(D)$. Conversely, take $x = \sum_i x_i \in D^\partial \cap \partial D$, where $x_i \in \partial_{L_i} D_i$. A supporting hyperplane at x has a nonzero normal u . Since support values add and each support deficit is nonnegative, every x_i maximizes u on its summand. Let $J = \{j : A_j^T u = 0\}$. For $i \notin J$, uniqueness gives $x_i = p_i(u)$.

All L_j , $j \in J$, lie in $u^\perp$. Condition (GP) forces their total dimension to be at most $d - 1$: otherwise their sum would have dimension d . It then gives equality between the dimension of their sum and the sum of their dimensions. Thus the concatenated matrix $A_J = [A_j]_{j \in J}$ is injective and A_J^T is surjective. Write $x_j = A_j v_j$ with $\|v_j\| = 1$; the relative boundary condition and injectivity justify this representation. Choose $w \in \mathbb{R}^d$ such that $A_j^T w = v_j$ for every $j \in J$.

If J is empty, x is already exposed. Otherwise, for $\varepsilon > 0$, set $u_\varepsilon = u + \varepsilon w$. Then

$$A_j^T u_\varepsilon = \varepsilon v_j, \quad p_j(u_\varepsilon) = A_j v_j = x_j \quad (j \in J).$$

For every remaining index, $A_i^T u_\varepsilon$ stays nonzero when ε is sufficiently small, and $p_i(u_\varepsilon) \rightarrow p_i(u) = x_i$. Finiteness gives one such bound for all indices. Consequently $u_\varepsilon \in U$ and $F(u_\varepsilon) \rightarrow x$. Every point generating $S(D)$ is a Euclidean limit of exposed points. Since the complex algebraic set $E(D)$ is Euclidean closed, $S(D) \subseteq E(D)$. Theorem 1 completes the proof. $\square$

For prescribed sizes $m_i \leq d$, (GP) is a nonempty real Zariski-open condition on the full-rank matrices A_i . Each failure of maximal concatenated rank is defined by minors. A simultaneous real witness is obtained by partitioning distinct Vandermonde columns $(1, t, \dots, t^{d-1})^T$ into blocks of sizes m_i . Every set of at most d columns is independent, and every larger set has rank d . When $\sum_i m_i \geq d$, this also ensures full dimensionality. This matches the bases-parametrized genericity convention in [1, Section 2]. As in that paper, a smaller ambient span may be replaced by the span of the discotope. Thus Theorem 4 proves its Conjecture 8.2 for every feasible generic type with no one-dimensional summands.

4 Why the hypotheses and the distinction matter

One-dimensional summands cannot be admitted indiscriminately. The interval $[-1, 1]$ has exposed-point closure $\{-1, 1\}$, which is reducible. The planar sum of the unit disc and $[-a, a]e_1$, $a > 0$, has exposed points on two open semicircles. Its complex Zariski closure is the union of the two distinct conics $(X - a)^2 + Y^2 = 1$ and $(X + a)^2 + Y^2 = 1$.

Nor does $S = E$ hold in arbitrary nongeneric position. In $\mathbb{R}^3$, take two identical unit discs in the xy -plane and one in the xz -plane. The point $e_3 = e_1 + (-e_1) + e_3$ belongs to $D^\partial \cap \partial D$, since the normal e_3 exposes the face $2B_{xy}^2 + e_3$. But $E(D)$ lies in the zero set of

$$P(X, Y, Z) = (X^2 + Y^2 + Z^2 - 5)^2 - 4(4 - Y^2)(1 - Z^2).$$

Indeed, for regular normals, write $r^2 = u_1^2 + u_2^2$, $s^2 = u_1^2 + u_3^2$, $a = 2u_1/r$, and $b = u_1/s$. Then $X = a + b$, $a^2 + Y^2 = 4$, and $b^2 + Z^2 = 1$, giving $X^2 + Y^2 + Z^2 - 5 = 2ab$ and $P = 0$. As $P(0, 0, 1) = 16$, we obtain $e_3 \in S(D) \setminus E(D)$. This is algebraic separation, not merely a failure of exposure by a selected normal.

Verification and use of AI

The accompanying verification package contains the manuscript source, exact symbolic checks with a pinned environment, source provenance, and independent adversarial reviews. The finite checks verify the separating polynomial, concrete general-position perturbations (including non-orthonormal ellipses), and boundary examples. They corroborate the formulas and do not replace the proofs of the universal theorems above.

Generative AI tools were used extensively for proof development, literature triage, manuscript preparation, adversarial mathematical review, and symbolic-check design. AI review is not human peer review or a formal proof certificate. This preprint is unrefereed. A dated priority audit is included in the package; historical first priority is not asserted.

References

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